

26 **J** is a synthetic nonapeptide that is found in honeybee venom.

To investigate the sequence of amino acids in **J**, the nonapeptide was first hydrolysed by two enzymes. The protein fragments were then separated and their sequence determined.

The first enzyme, which hydrolysed the polypeptide chain at the carboxylic end of the amino acid lysine, Lys, yielded the following fragments.

Trp-Ile-Lys
Leu-Arg
Arg-Ile-Ser-Lys

The following protein fragments were obtained from the second enzyme which hydrolysed the polypeptide chain at the carboxylic end of the amino acid isoleucine, Ile.

Ser-Lys-Trp-Ile
Arg-Ile
Lys-Leu-Arg

Which is the correct primary structure of the nonapeptide **J**?

- A** Lys-Leu-Arg-Ile-Ser-Lys-Trp-Ile-Lys
- B** Arg-Ile-Ser-Lys-Trp-Ile-Lys-Leu-Arg
- C** Ser-Lys-Trp-Ile-Lys-Leu-Arg-Ile-Ser
- D** Trp-Ile-Lys-Leu-Arg-Ile-Ser-Lys-Trp